

The exact image size of every quadratic translate over a prime field

CONJECTURE 00000000075 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and formalization

Conjecture clause

The image of $x \mapsto x^2 + c$ on the prime field F_p is asymptotic to $(1 - \sqrt{\pi/8}) \cdot p$. Refuting this clause refutes the conjunction in the filed conjecture.

Exact counting

For odd prime p , equality $x^2 = y^2$ is equivalent to $x = y$ or $x = -y$. Zero has one square root; every nonzero square has two distinct square roots. Thus the square map has $1 + (p-1)/2 = (p+1)/2$ distinct images. Adding c is a bijection, so this count is unchanged for every c .

Asymptotic contradiction

The normalized image size is exactly $1/2 + 1/(2p)$, and tends to $1/2$ along odd primes. The proposed coefficient $1 - \sqrt{\pi/8}$ is less than $1/2$, because $\pi > 3 > 2$. A sequence all of whose terms are at least $1/2$ cannot converge to that smaller coefficient.

Scope

The argument works for every c , so exceptional parameters do not repair the image-size clause. It makes no claim about the separate cyclic-point and tail-depth assertions. Lean proves the exact cardinality and the impossibility of the conjectured limiting coefficient for any sequence of odd-prime instances.

Reproduction

Pinned Lean and Mathlib: v4.33.0. In lean4/: lake exe cache get; lake build; lake env lean Check.lean. The independent reproduce.py checks finite instances; the written general proof and Lean theorems settle the infinite statements.

Source: repository conjectures/00000000075.md; main branch df87a69ed4814eeb34ad161879b53db10c44fdc6.